MATHUSAN MANIVANNAN

Data Science Undergraduate | Aspiring Data Engineer

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PROFESSIONAL SUMMARY

Data Science undergraduate and aspiring data engineer who builds end-to-end pipelines with Python, SQL, PySpark, Airflow, and Azure. Co-founder of a student software startup with a production client system delivered. Seeking a Data Engineering Internship to build reliable, scalable data pipelines.

EDUCATION

BSc (Hons) in Information Technology, Specialization in Data Science 2024 - Present

Sri Lanka Institute of Information Technology (SLIIT)

PROJECTS

E-Commerce Sales ETL Pipeline *August - 2026*

Python, Pandas, PostgreSQL, SQL, Apache Airflow, Docker

- Developed an end-to-end ETL pipeline to extract and transform e-commerce sales data using Python and Pandas.
- Implemented data cleaning, validation, deduplication and business-rule transformations before loading processed data into PostgreSQL.
- Designed a star-schema data warehouse with fact and dimension tables for analytical querying.
- Automated ETL workflow execution using Apache Airflow with task dependencies, scheduling and retry handling.
- Containerized the development environment using Docker.

Azure End-to-End Data Engineering Pipeline (Medallion Architecture) *Sep 2026*

Azure Data Factory, Databricks (PySpark), ADLS Gen2, Azure Synapse Analytics, Power BI

- Built an end-to-end Azure data platform on the AdventureWorks dataset (**10 source files, 3 years of sales data**) using a Bronze/Silver/Gold medallion architecture on ADLS Gen2.
- Developed a parameterized, metadata-driven ADF pipeline (Lookup + ForEach + JSON config, **3 dynamic parameters**) that ingests all **10 files** from a GitHub HTTP source through **1 reusable dataset**.
- Transformed data in Databricks with PySpark (date, string and regex functions, groupBy aggregations), wrote **8 tables** to the Silver layer as Parquet, and secured storage access with a service principal and managed identity (RBAC).
- Served the Gold layer via Synapse serverless SQL (OPENROWSET views, CETAS external tables in Parquet) and connected Power BI to build sales trend and KPI visuals on data-lake data.

Global Energy Consumption & Electricity Demand Analytics *Oct 2026*
Python, SQL Server, Power BI

- Built an end-to-end ETL pipeline using Python/Pandas across 3 data sources, processing 23K+ energy records and data covering 200+ countries and 100+ years.
- Designed a SQL Server Data Warehouse & Energy Data Mart with fact/dimension tables for analytical querying and reporting.
- Built Year/Decade hierarchies and drill-down measures to analyze long-term energy consumption and electricity trends across countries.
- Developed an interactive Power BI dashboard with KPIs, country comparisons, heatmaps, trends, and contextual energy insights.

Full-Stack POS & Student Order System *May 2026*
Python, React, PostgreSQL | Client Project

- Designed and implemented a PostgreSQL-backed POS and student order management system for a real client
- Developed data models and database relationships for products, orders, users, payments, and transaction record.
- Built backend APIs to support reliable data creation, retrieval, updating, and reporting workflows
- Participated in requirements gathering, development, testing, deployment, and client handover.

RECENT CERTIFICATES

- Python for Beginners - UOM
- Data Analysis – Microsoft Simplelearn

TECHNICAL SKILLS

Programming: Python, SQL, PySpark,

Data Engineering: ETL/ELT, Data Cleaning, Data Validation, Data Pipelines, Data Warehousing, Batch Processing, Dimensional Modeling

Cloud: Azure Data Factory, Azure Databricks, Azure Synapse Analytics, ADLS Gen 2, Apache Airflow

Data & Analytics: Pandas, NumPy

Databases: MySQL, PostgreSQL,

Tools: Git, GitHub, Docker, Power BI

REFERENCES

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